DESIGN AND 3D MODELING OF A MECHANICAL GEARBOX ASSEMBLY USING AUTOCAD

Submitted By:

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Internship:

CodeAlpha AutoCAD Internship

Software Used:

AutoCAD 2026

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Objective

The objective of this project is to create a complete 3D mechanical assembly using AutoCAD. The project focuses on developing practical skills in 3D solid modeling, assembly creation, and the use of AutoCAD tools to produce a realistic engineering model.

Introduction

AutoCAD is a powerful computer-aided design software widely used for creating engineering drawings and 3D models. In this project, a mechanical gearbox assembly was modeled using AutoCAD 2026 and solid modeling techniques. Various features such as cylindrical bodies, flanges, holes, and fillets were created and combined to form the final assembly. This project helped in improving practical knowledge of 3D modeling workflows and engineering visualization.

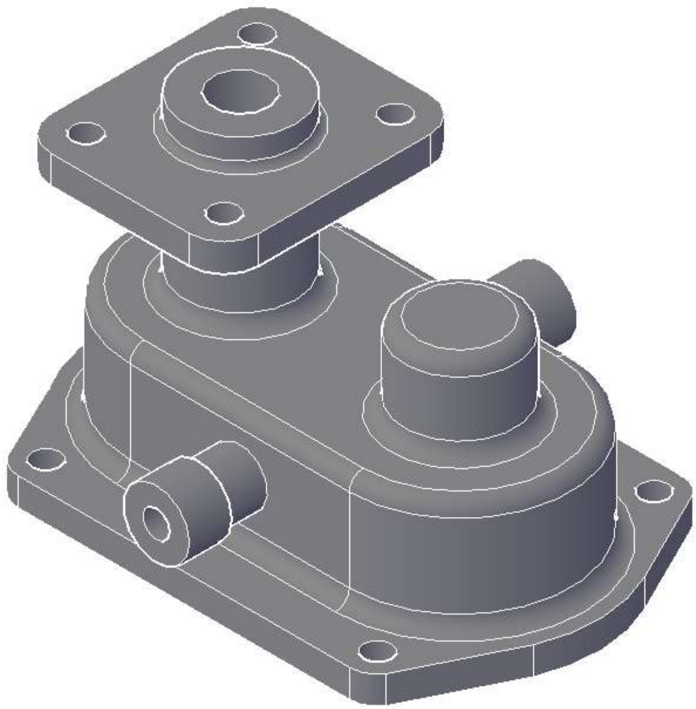

Figure 1: Isometric View of the Mechanical Gearbox Assembly

DESIGN WORKFLOW

1. Created the basic sketches using circles, rectangles, and reference geometry.
2. Converted the 2D sketches into 3D solid models using the EXTRUDE command.
3. Combined different components into a single assembly using the UNION command.
4. Generated holes and internal features using the SUBTRACT command.
5. Applied smooth edges and finishing details using the FILLETEDGE command.
6. Positioned and aligned components accurately using MOVE and ROTATE.
7. Verified the final assembly and generated different orthographic and isometric views for documentation.

AUTOCAD COMMANDS USED

• LINE

• CIRCLE

• RECTANGLE

• EXTRUDE

• UNION

• SUBTRACT

• FILLETEDGE

• MOVE

• ROTATE

• 3DORBIT

• ZOOM EXTENTS

COMPONENTS OF THE ASSEMBLY

- Base Plate
- Main Housing
- Vertical Support Shaft
- Top Flange Plate
- Side Cylindrical Connectors
- Auxiliary Cylindrical Chamber
- Mounting Holes

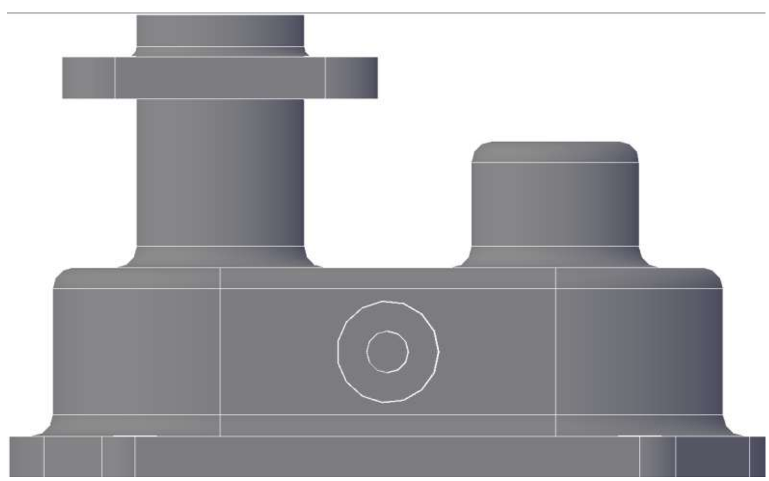

Figure 2: Front View of the Mechanical Gearbox Assembly

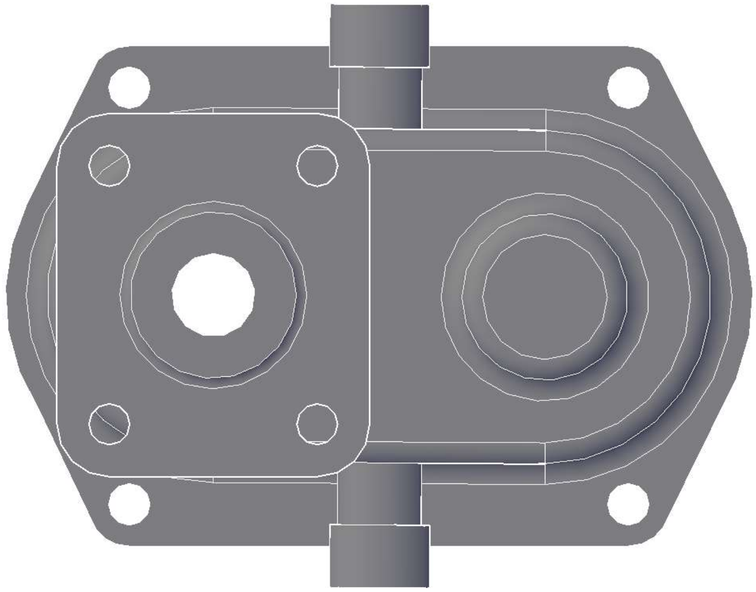

Figure 3: Top View of the Mechanical Gearbox Assembly

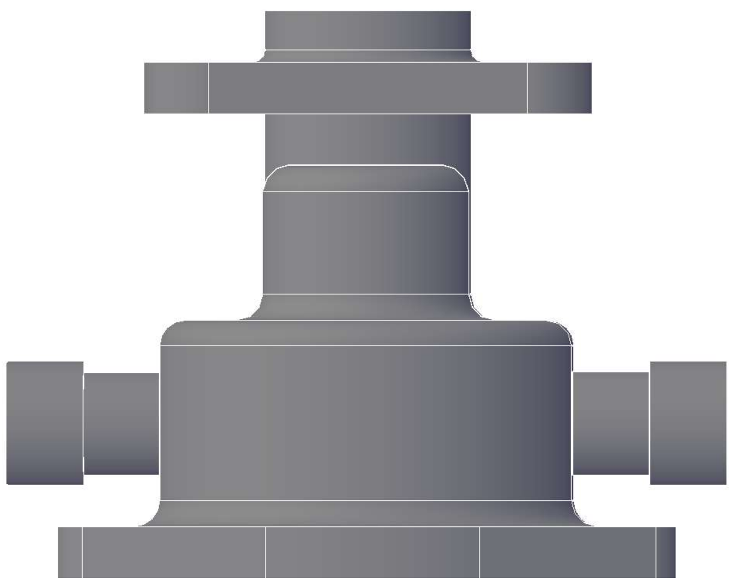

Figure 4: Side View of the Mechanical Gearbox Assembly

CONCLUSION

This project provided practical experience in 3D solid modeling using AutoCAD 2026. Various commands such as Extrude, Union, Subtract, and Fillet Edge were used to create a complete mechanical gearbox assembly. The project improved understanding of engineering visualization, assembly design, and CAD documentation techniques.